

Science Portal & API

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On behalf of Fink

cnrs

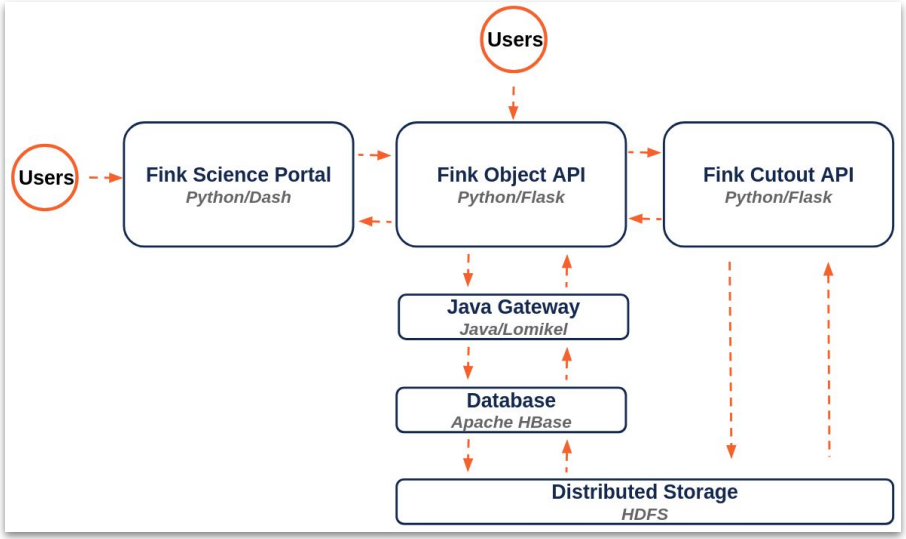
Infrastructure

Services deployed on the VirtualData cloud @ UPSaclay (France)

- Distributed database (HBase)
  - +200M rows, 3TB
- Cutout storage service
  - 1B cutouts, 8TB
- Web services
  - REST API, science portal

Anonymous usage, no limits

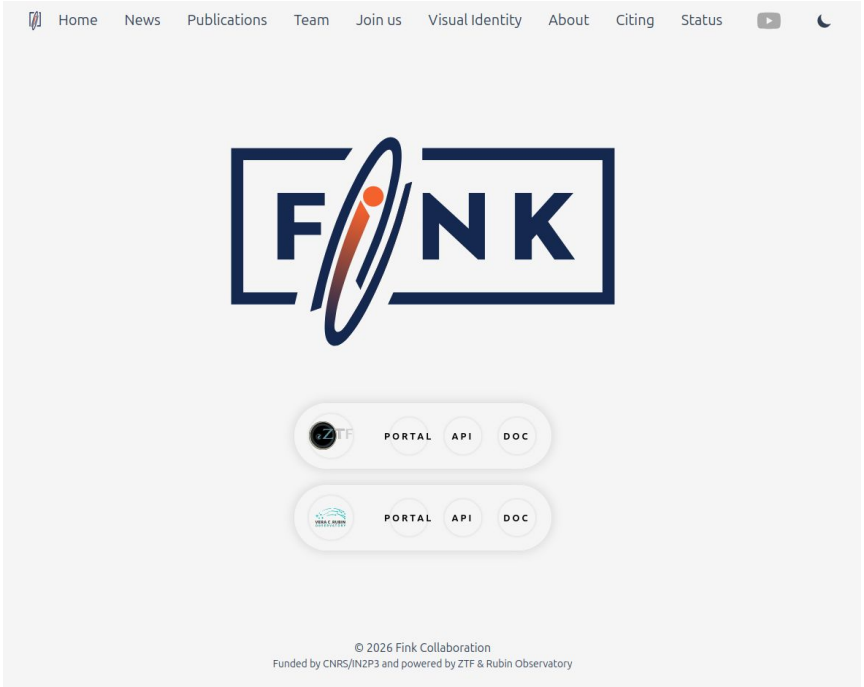
About 50k requests/day.



Open-source code on GitHub:
astrolabsoftware/fink-object-api
astrolabsoftware/fink-cutout-api
astrolabsoftware/fink-science-portal

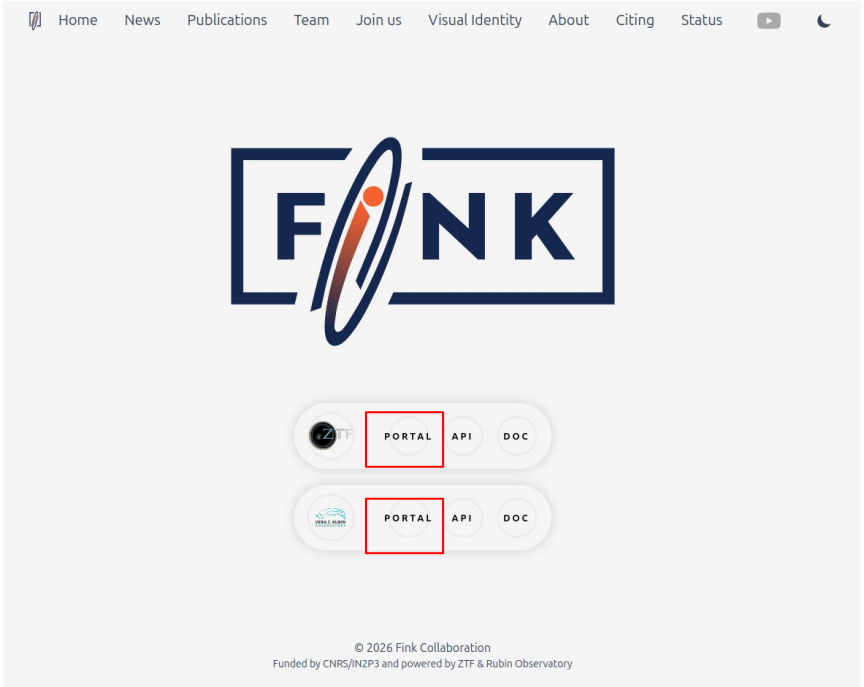
Point of entry

https://fink-broker.org/



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Science Portal

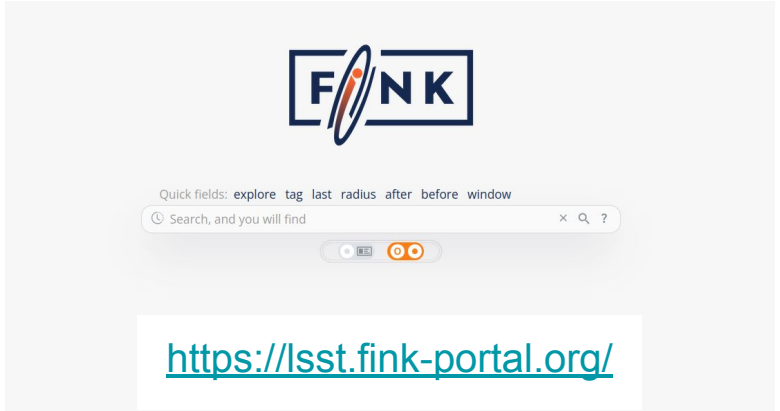
What it does allow

- Enable simple queries on name, position or alert class
- Let you visualise and interact with the results on the browser



What it does not allow

- Perform complex analyses
  - See Data Transfer or Livestream
- Manage follow-up
  - But connection to TOM/Marshalls possible



Search

Quick search based on

- **Name**
  - `NGC 710 r=30`
  - `SN 2025zmx`
  - `ZTF25abwjtwc / 313831458352922664`
  - `sso=Melbourne`
- **Position**
  - 01 52 54.12 +36 03 19.5
  - 01h52m54.12s +36d03m19.37s
  - 28.225482 36.055381
  - 28.225482 36.055381 r=60
- **Class (ZTF) / Tag (LSST)**
  - class="Early SN Ia candidate"
  - tag="extragalactic_new_candidate"

Card view

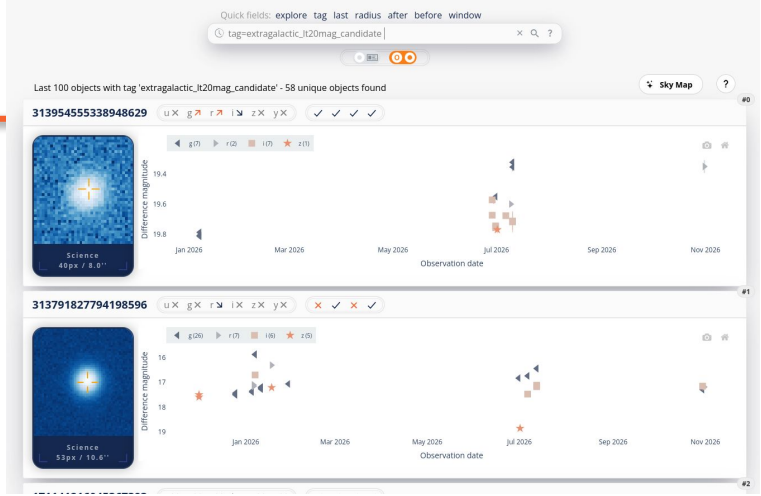
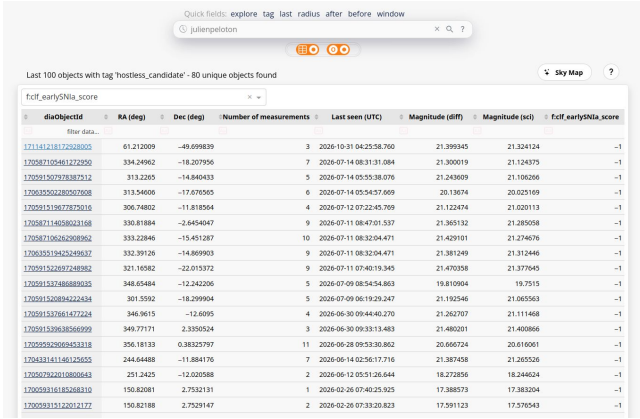
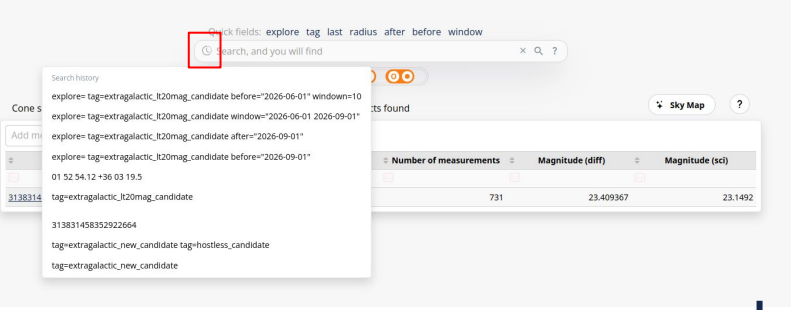


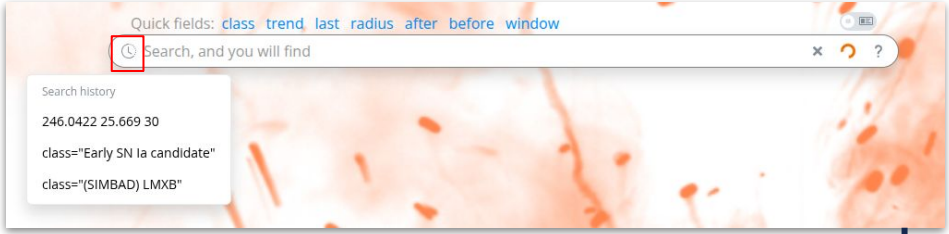
Table view



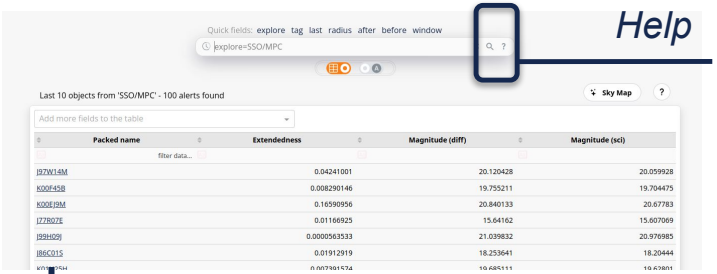
Search bar tips



Search history

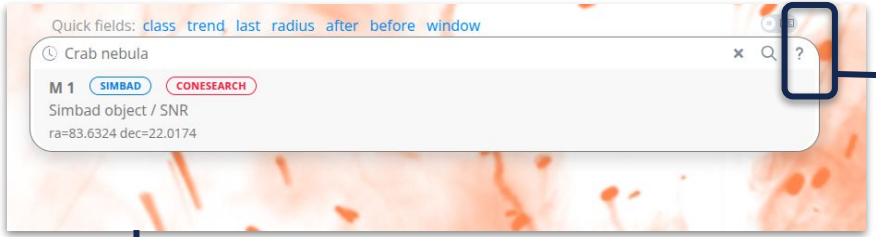


Search history



Help

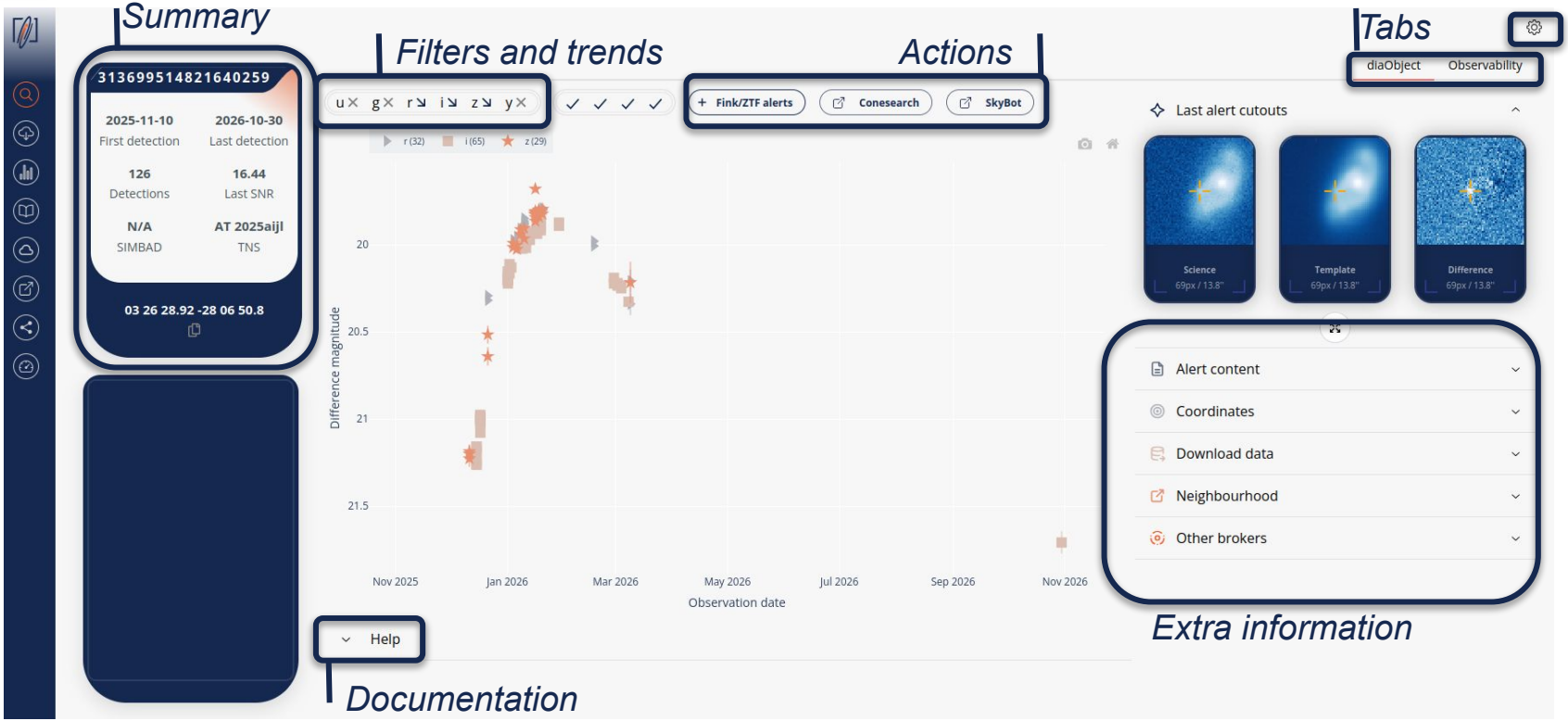
Query



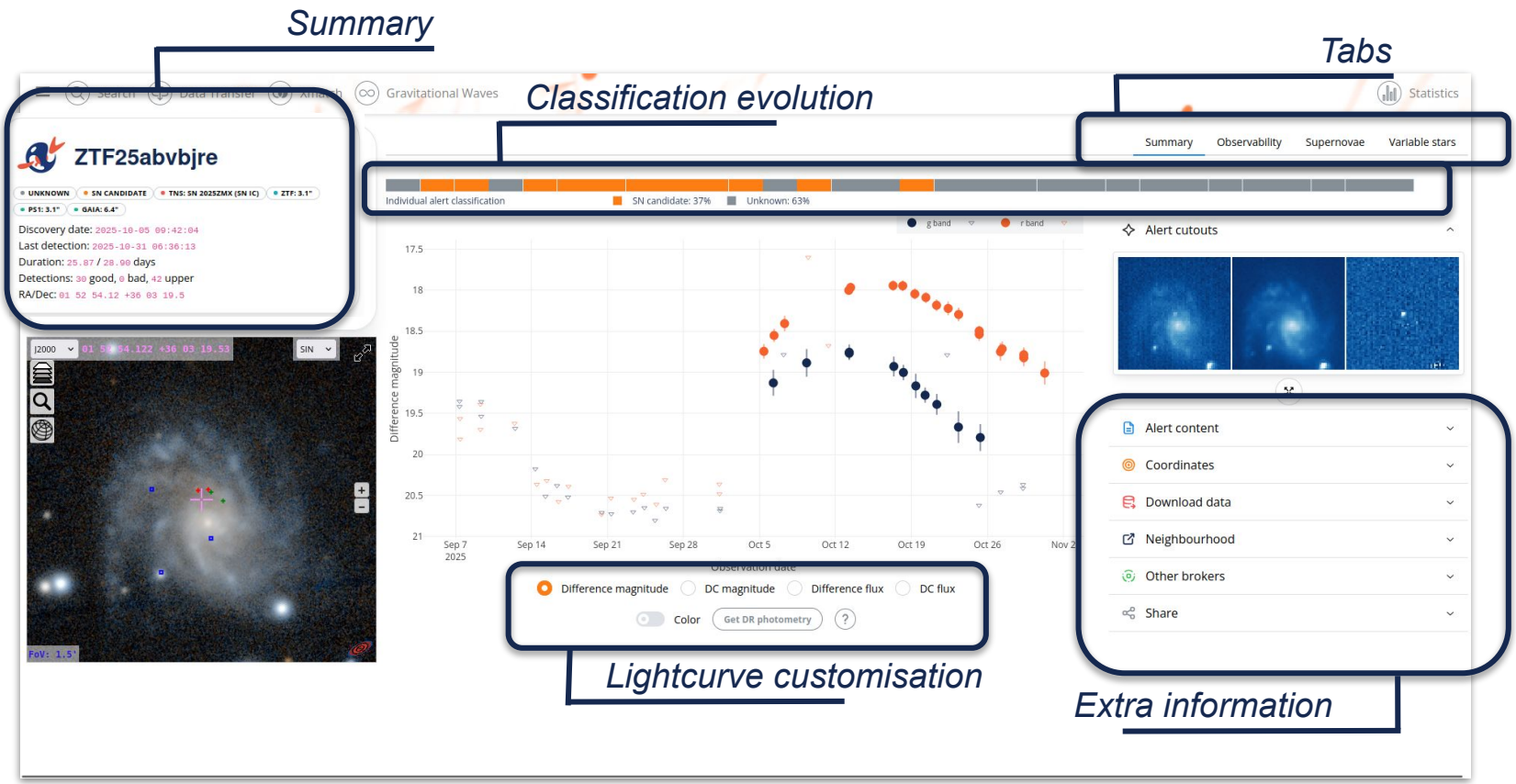
Help

Resolving query

Object page - LSST



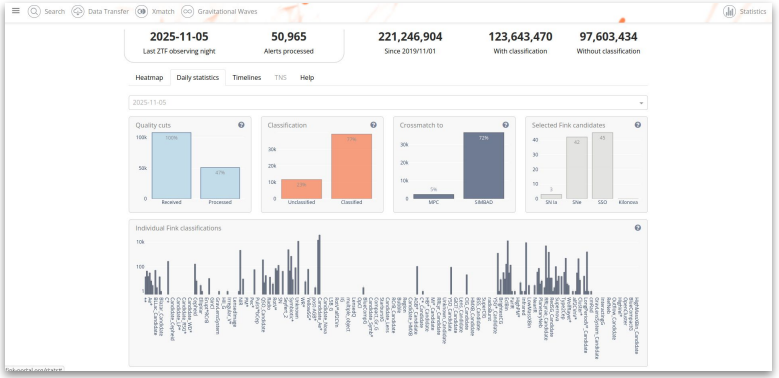
Object page - ZTF

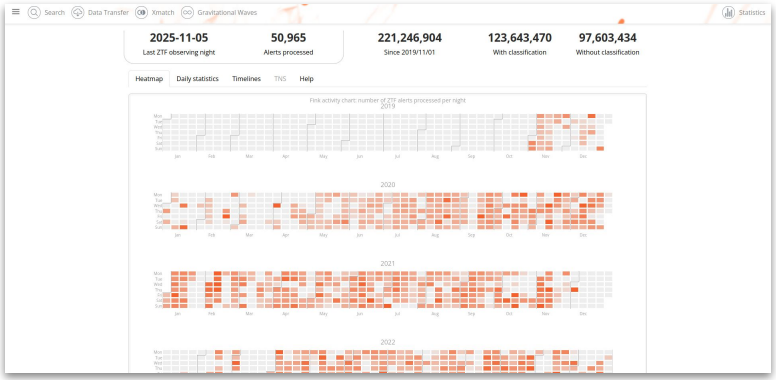


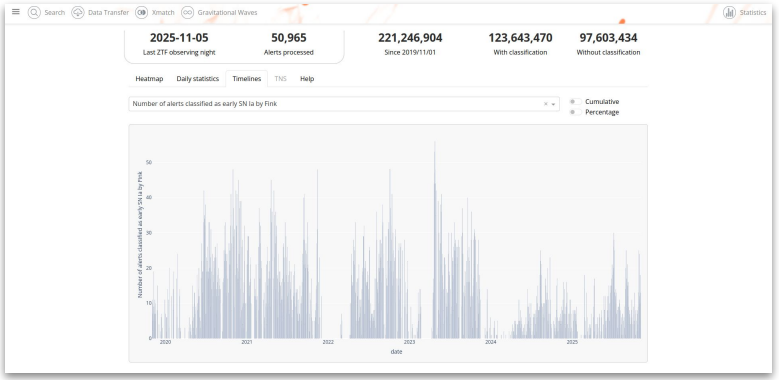
Statistics

Various statistics on the survey, and Fink outputs

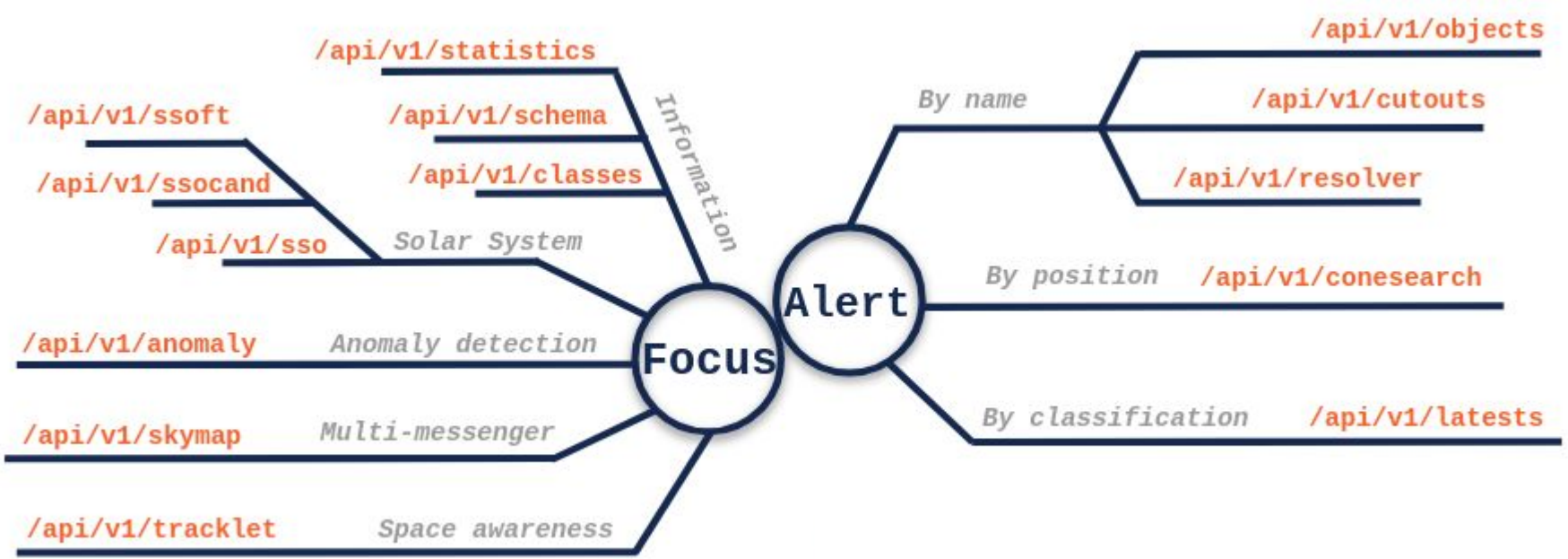
- Per-night performances
- Classification evolution







API



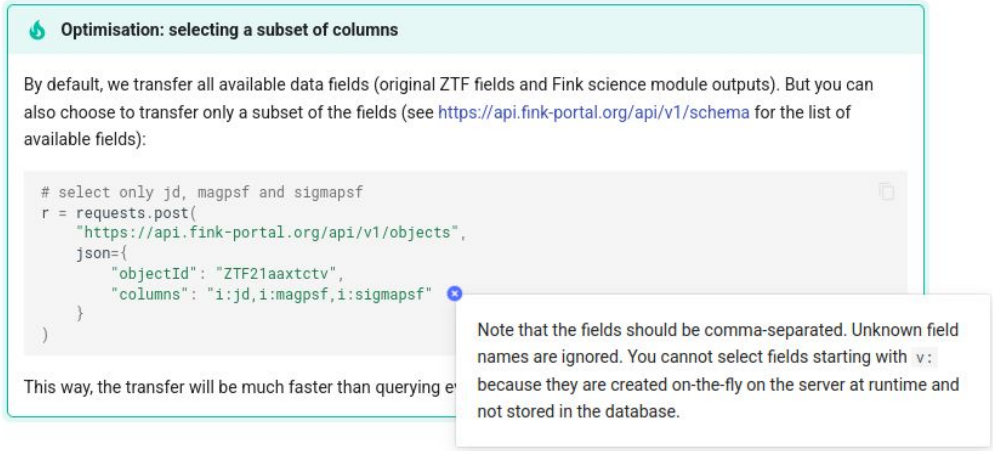
ZTF: https://api.ztf.fink-portal.org/
LSST: https://api.lsst.fink-portal.org/

Tips and trick

#1 Select only columns you need!

#2 Results are cached. Hence if you hit a timeout, relaunched the query, and you should get results the next time.

#3 You can run queries on the URL, e.g.



https://ztf.fink-portal.org/?action=conesearch&ra=193.822&dec=2.89732&r=5
https://ztf.fink-portal.org/?action=class&class=Early SN Ia candidate

Bookmark your favourite ones, to visualise results in one click everyday!

Using the API

Extragalactic tutorial:

https://drive.google.com/file/d/1qyGOXzYnLr6hnb9djWqhG1-6wTvxUdTs/view?usp=sharing